

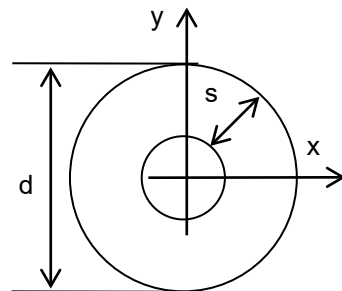
Fundamentals of Strength of Materials – 2018-2019

Tutorial #7: Torsion

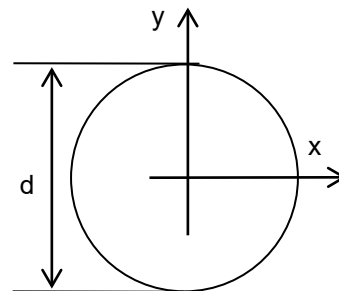
Example #1

To the solid and hollow circular sections shown in following figure are applied a bending moment $M_x=750$ Nm and a torque $M_z=400$ Nm, respectively. Relevant dimensions are $d=30$ mm and $s=5$ mm.

For each section draw the distribution of normal (σ) and shear (τ) stresses, by indicating the point where the stress is maximum. Compute the safety factor in case of a) ductile material (S460 - $R_{eH} = S_y = 460$ MPa) and b) brittle material (GGG60 - $R_m = S_{UTS} = 600$ MPa)



a) $F_S = 1.1$; b) $F_S = 1.6$

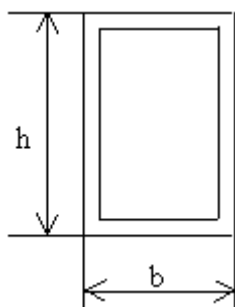


a) $F_S = 1.4$; b) $F_S = 2$

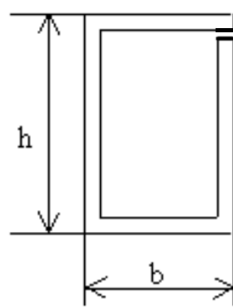
Example #2

The thin-walled sections shown in following figure undergo the effect of torque $M_z=20$ Nm. Draw the tangential stress distribution and compute the maximum value.

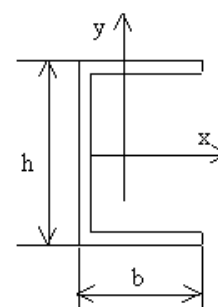
$d = 30$ mm, $h=30$ mm, $b=20$ mm, thickness $t=2$ mm.



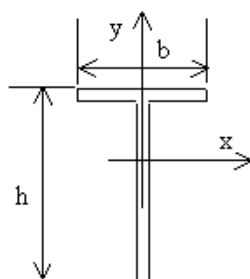
$\tau_{\max} = 10$ MPa



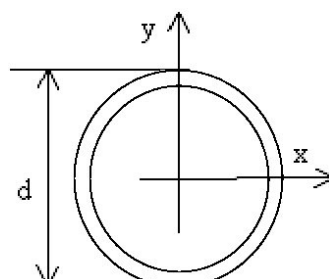
$\tau_{\max} = 163$ MPa



$\tau_{\max} = 227$ MPa



$\tau_{\max} = 305$ MPa



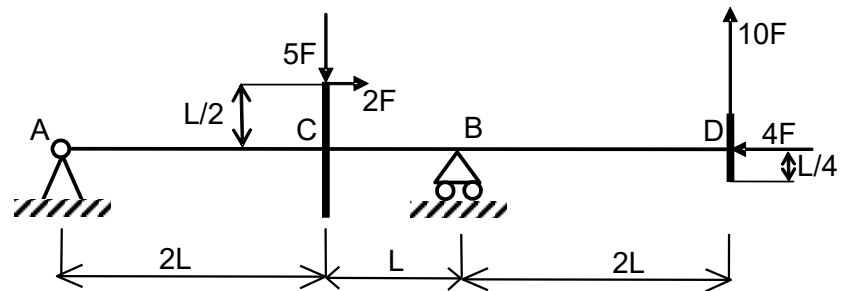
In this case use first the beam theory applied to the hollow circular section (a) then the Bredt's theory for thin walled closed sections (b).

a) $\tau_{\max} = 9$ MPa; b) $\tau_{\max} = 8$ MPa

Example #3

The cross section of the shaft sketched in figure is circular and has diameter 30 mm, being 36CrNiMo4 steel ($R_{eH} = S_y = \sigma_y = 800 \text{ MPa}$). Loads are shown in figure and in addition a constant torque of 600 Nm is applied on the shaft, between the two gears. Assume $F = 500 \text{ N}$ and $L = 100 \text{ mm}$.

Compute the safety factor against yielding and the relative rotation of sections C-B and C-D.



$$F_S = 1.8; \varphi_{CB} = 0.56^\circ; \varphi_{CD} = 1.68^\circ;$$